

EXHIBIT 27

Appl. No. 13/550,571
Amdt dated January 25, 2013
Reply to Office Action dated October 26, 2012

PATENT

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Appl. No. : 13/550,571 Confirmation No. : 1089
Applicant : Guo Wei ZHANG et al.
Filed : 07/16/2012
TC / A.U. : 2892
Examiner : HOANG, QUOC DINH
Docket No. : CSM P 2009 NAT 03 US 2
Customer No. : 016162
Title : HIGH VOLTAGE DEVICE

CERTIFICATE OF TRANSMISSION

I hereby certify that this correspondence is being electronically
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Signed: _____

/mei sin tan/

Mei Sin TAN

Mail Stop AMENDMENT
Commissioner For Patents
P.O. Box 1450
Alexandria, VA 22313-1450

AMENDMENT

Sir:

In response to the Office Action mailed on October 26, 2012, please amend the above-identified application as follows:

- a) **Amendments to the Claims** are reflected in the listing of claims which begins on page 2 of this paper; and
- b) **Remarks/Arguments** begin on page 7 of this paper.

Amendments to the Claims

The listing of claims will replace all prior versions, and listings, of claims in the application:

Listing of Claims:

1. (currently amended) A device comprising:

[[s]]a substrate prepared with an active device region, wherein the active device region includes a gate, the gate including a gate electrode over a gate dielectric;

a doped channel well disposed in the substrate adjacent to a first edge of the gate, the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge beneath the gate defines a an ~~effective channel length~~ of the device, wherein the ~~effective channel length is self-aligned to the~~ channel is displaced from a first heavily doped region adjacent to the first edge of the gate; and

a doped drift well adjacent to a second edge of the gate.

2. (currently amended) The device of claim 1 ~~wherein the effective channel length is less than about 0.4 μ m~~ further comprising spacer on the first edge of the gate, wherein the spacer defines the displacement of the channel from the first heavily doped region.

3. (original) The device of claim 1 wherein the substrate comprises:

a drift isolation region in the doped drift well; and

a doped deep well in the substrate of the active device region encompassing the doped drift well.

4. (original) The device of claim 1 wherein the first edge of the gate corresponds to a channel edge of the gate of the device and a second edge of the gate corresponds to a drain edge of the gate of the device.
5. (original) The device of claim 3 wherein the drift isolation region is disposed under the gate.
6. (currently amended) The device of claim 5 wherein the first and a second ~~diffusion~~ heavily doped regions are provided adjacent to the gate, wherein:
the first ~~diffusion~~ heavily doped region is disposed in the doped channel well on a source side of the device; and
the second ~~diffusion~~ heavily doped region is disposed between the drift isolation region and a device isolation region on a drain side of the device.
7. (currently amended) The device of claim 1 wherein the ~~effective~~ channel having a channel length that is controlled by ion implantation.
8. (original) The device of claim 7 wherein the ion implantation comprises a tilt angle implant.
9. (original) The device of claim 8 wherein a tilt angle of the ion implantation is about 1-45°.
10. (original) The device of claim 7 wherein the ion implantation comprises multiple tilt angle implants which are rotated about a plane of the substrate.

11. (original) The device of claim 10 wherein a tilt angle of the implant is about 1-45°.

12. (original) The device of claim 7 wherein the ion implantation comprises a quad tilt angle implant which is rotated about a plane of the substrate.

13. (currently amended) A device comprising:

[[s]]a substrate prepared with an active device region, wherein the active device region includes a gate, the gate including a gate electrode over a gate dielectric;

a doped channel well disposed in the substrate adjacent to a first edge of the gate, wherein the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge beneath the gate defines a an effective-channel length of the device, wherein the ~~effective-channel length~~ is self-aligned to the gate channel is displaced from a first heavily doped region adjacent to the first edge of the gate;

a doped drift well adjacent to a second edge of the gate; and

a drift isolation region in the doped drift well.

14. (currently amended) The device of claim 13 ~~wherein the effective-channel length is less than about 0.4 μm~~ further comprising a spacer on the first edge of the gate, wherein the spacer defines the displacement of the channel from the first heavily doped region.

15. (original) The device of claim 13 wherein the first edge of the gate corresponds to a channel edge of the gate of the device and a second edge of the gate corresponds to a drain edge of the gate of the device.

16. (currently amended) The device of claim 13 wherein the drift isolation region is disposed under the gate.

17. (currently amended) The device of claim 13 wherein the first and a second ~~diffusion~~ heavily doped regions are provided adjacent to the gate, wherein:

the first ~~diffusion~~ heavily doped region is disposed in the doped channel well on a source side of the device; and

the second ~~diffusion~~ heavily doped region is disposed between the drift isolation region and a device isolation region on a drain side of the device.

18. (currently amended) The device of claim 13 wherein the ~~effective~~ channel having a channel length that is controlled by ion implantation.

19. (original) The device of claim 18 wherein the ion implantation comprises a tilt angle implant, multiple tilt angle implants or quad tilt angle implant.

20. (currently amended) A semiconductor device comprising:

[[s]]a substrate prepared with an active device region, wherein the active device region includes a gate, the gate including a gate electrode over a gate dielectric;

first and second heavily doped regions adjacent to first and second edges of the gate;

a doped channel well disposed in the substrate adjacent to [[a]] the first edge of the gate, wherein the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge beneath the gate defines a an effective channel length of the device, wherein the effective channel length is self-aligned to the gate channel is displaced from the first heavily doped region adjacent to the first edge of the gate; and

a doped drift well adjacent to a second edge of the gate, wherein a drift isolation region is disposed in the doped drift well; ~~and~~

~~first and second diffusion regions adjacent to the gate.~~

Remarks/Arguments

Claim Amendments

Without conceding the propriety of the rejections herein and in the interest of expediting prosecution, Applicants amend some claims herein. Applicants amend claims to clarify claimed features. Such amendments are made to expedite prosecution and more quickly identify allowable subject matter. Such amendments are merely intended to clarify the claimed features, and should not be construed as further limiting the claimed invention in response to the cited references. All amendments are fully supported by the Application and therefore do not constitute new matter.

Claim Rejections – 35 USC § 102

Claims 1, 4, and 7-12 are rejected under 35 USC § 102(b) as being anticipated by US Patent No. 6,979,864 (Negoro). Applicants respectfully disagree with this basis of rejection.

With respect to independent claim 1, Applicants respectfully submit that Negoro fails to disclose at least the channel is displaced from a first heavily doped region adjacent to the first edge of the gate. In contrast, Negoro discloses the surface of the p-type channel well region (element 23) under the gate electrode (element 11g) is operative as a channel region. The inner edge of the n-type heavily doped region (element 11s) and the inner edge of the p-type channel well define the channel. Therefore, Negoro fails to disclose that the channel is displaced from the n-type heavily doped region. *See, e.g., Negoro*, column 15, lines 19-21; FIG. 1.

Claims 4 and 7-12 ultimately depend upon independent claim 1. As discussed above, claim 1 is allowable. It is axiomatic that any dependent claim which depends from an

allowable base claim is also allowable. Additionally, some or all of these claims may also be allowable for additional independent reasons.

Consequently, Negoro fails to disclose “each and every element” as set forth in these claims. Accordingly, Applicants ask the Examiner to withdraw the rejection of these claims.

Claim Rejections – 35 USC § 103

Claim 2 is rejected under 35 USC § 103(a) as being unpatentable over Negoro in view of US Patent No. 7,829,402 (Matocha). Applicants respectfully disagree with this basis of rejection.

Claim 2 ultimately depends on independent claim 1. As already discussed, independent claim 1 is patentable over Negoro. In particular, Negoro fails to disclose at least the channel is displaced from a first heavily doped region adjacent to the first edge of the gate. To compensate for the defects of Negoro, the Examiner relies on Matocha. Matocha, like Negoro, fails to teach or suggest these features of the independent claim 1. This is evidenced by the fact that the Examiner only relies on Matocha to teach the channel length. Applicants therefore submit that independent claim 1, as well as its directly or indirectly dependent claims, including claim 2, are patentable over Negoro and Matocha, alone or in combination.

Claims 3, 5-6, 13, and 15-20 are rejected under 35 USC § 103(a) as being unpatentable over Negoro in view of US Patent No. 7,935,992 (Chang). Applicants respectfully disagree with this basis of rejection.

Claims 3 and 5-6 ultimately depend on independent claim 1. As already discussed, independent claim 1 is patentable over Negoro. It is axiomatic that any dependent claim which depends from an allowable base claim is also allowable. Additionally, these claims may also be allowable for additional independent reasons. Accordingly, Applicants

respectfully submit that this basis of rejection has been traversed and respectfully request its withdrawal.

With respect to independent claims 13 and 20, these claims generally recite a device. Claim 13 is representative. This claim recites a device which includes a substrate prepared with an active device region. The active device region includes a gate. The gate includes a gate electrode over a gate dielectric. The device also includes a doped channel well. The doped channel well is disposed in the substrate adjacent to a first edge of the gate. The first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate. The first edge of the gate and channel edge beneath the gate define a channel of the device. The channel is displaced from a first heavily doped region adjacent to the first edge of the gate. The device also includes a doped drift well adjacent to a second edge of the gate and a drift isolation region in the doped drift well.

In the Office Action (page 5), the Examiner admits that Negoro fails to teach a drift isolation region in the doped drift well and relies on Chang to compensate for its defects. However, the addition of Chang fails to compensate for the defects of Negoro. For example, Chang also fails to teach or suggest at least the channel is displaced from a first heavily doped region adjacent to the first edge of the gate. Therefore, the combined teaching of Negoro and Chang at most, teaches or suggests a device having a channel defines by the edge of the heavily doped (source/drain) region and the edge of the channel well. As such, one of ordinary skilled in the art would not be led to forming a device having a channel which is displaced from the edge of a first heavily doped region adjacent to the first edge of the gate as recited in the present claims.

Accordingly, for at least the reasons discussed above, the asserted combination of Negoro and Chang does not teach or suggest all of the elements and features of these claims. Applicants submit that independent claims 13 and 20 as well as their directly and indirectly

dependent claims are patentable over Negoro and Chang, alone or in combination. Accordingly, Applicants respectfully request the Examiner to withdraw the rejection of these claims.

Conclusion

It is respectfully submitted that this application is in condition for allowance and the issuance of a formal Notice of Allowance at an early date is respectfully requested.

Should the Examiner believe that a telephone conference would expedite prosecution of this application, please telephone the undersigned attorney at his number set out below.

Date: January 25, 2013

Respectfully submitted,

/dexter chin/

Dexter CHIN
Attorney for Applicants
Reg. No.: 38,842

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TRANSMITTAL FORM <i>(to be used for all correspondence after initial filing)</i>	Application Number	13/550,571	
	Filing Date	07/16/2012	
	First Named Inventor	Guo Wei ZHANG	
	Art Unit	2892	
	Examiner Name	HOANG, QUOC DINH	
Total Number of Pages in This Submission	12	Attorney Docket Number	CSM P 2009 NAT 03 US 2

ENCLOSURES (Check all that apply)		
☐ Fee Transmittal Form ☐ Fee Attached ☒ Amendment/Reply ☐ After Final ☐ Affidavits/declaration(s) ☐ Extension of Time Request ☐ Express Abandonment Request ☐ Information Disclosure Statement ☐ Certified Copy of Priority Document(s) ☐ Reply to Missing Parts/ ☐ Reply to Missing Parts under 37 CFR 1.52 or 1.53	☐ Drawing(s) ☐ Licensing-related Papers ☐ Petition ☐ Petition to Convert to a Provisional Application ☐ Power of Attorney, Revocation ☐ Change of Correspondence Address ☐ Terminal Disclaimer ☐ Request for Refund ☐ CD, Number of CD(s) _____ ☐ Landscape Table on CD	☐ After Allowance Communication to TC ☐ Appeal Communication to Board of Appeals and Interferences ☐ Appeal Communication to TC (Appeal Notice, Brief, Reply Brief) ☐ Proprietary Information ☐ Status Letter ☐ Other Enclosure(s) (please identify below):
Remarks		

SIGNATURE OF APPLICANT, ATTORNEY, OR AGENT			
Firm Name	Horizon IP Pte. Ltd.		
Signature	/dexter chin/		
Printed name	Dexter CHIN		
Date	01/25/2013	Reg. No.	38,842

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Typed or printed name	Yi Jun LAW	Date	01/25/2013

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9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Acknowledgement Receipt

EFS ID:	14788221
Application Number:	13550571
International Application Number:	
Confirmation Number:	1089
Title of Invention:	HIGH VOLTAGE DEVICE
First Named Inventor/Applicant Name:	Guo Wei ZHANG
Customer Number:	16162
Filer:	Dexter Kam Chin
Filer Authorized By:	
Attorney Docket Number:	CSM P 2009 NAT 03 US 2
Receipt Date:	25-JAN-2013
Filing Date:	16-JUL-2012
Time Stamp:	01:31:11
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Amendment/Req. Reconsideration-After Non-Final Reject	CSMP2009NAT03US2-ResponsetoNonFinalOA.pdf	130847 2d5ba3459f280ccb9a445a5a0183221be16014	no	10

Warnings:

Information:

2	Miscellaneous Incoming Letter	CSMP2009NAT03US2-TF.pdf	294013	no	2
			45de55e876740d6b9daa01b30115650b1e b39834		
Warnings:					
Information:					
Total Files Size (in bytes):				424860	
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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875					Application or Docket Number 13/550,571		Filing Date 07/16/2012		☐ To be Mailed				
APPLICATION AS FILED – PART I													
(Column 1)			(Column 2)			SMALL ENTITY ☐ OR		OTHER THAN SMALL ENTITY					
FOR		NUMBER FILED		NUMBER EXTRA		RATE (\$)		FEE (\$)					
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))		N/A		N/A		N/A							
☐ SEARCH FEE (37 CFR 1.16(k), (l), or (m))		N/A		N/A		N/A							
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))		N/A		N/A		N/A							
TOTAL CLAIMS (37 CFR 1.16(i))		minus 20 =		*		X \$ =		OR		X \$ =			
INDEPENDENT CLAIMS (37 CFR 1.16(h))		minus 3 =		*		X \$ =		OR		X \$ =			
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$250 (\$125 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).											
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))													
* If the difference in column 1 is less than zero, enter "0" in column 2.													
APPLICATION AS AMENDED – PART II													
(Column 1)			(Column 2)			(Column 3)			SMALL ENTITY OR		OTHER THAN SMALL ENTITY		
AMENDMENT	01/25/2013		CLAIMS REMAINING AFTER AMENDMENT				HIGHEST NUMBER PREVIOUSLY PAID FOR		PRESENT EXTRA				
	Total (37 CFR 1.16(i))		* 20		Minus		** 20		= 0				
	Independent (37 CFR 1.16(h))		* 3		Minus		*** 3		= 0				
	☐ Application Size Fee (37 CFR 1.16(s))												
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))												
RATE (\$)		ADDITIONAL FEE (\$)				RATE (\$)		ADDITIONAL FEE (\$)					
X \$ =						OR		X \$62=		0			
X \$ =						OR		X \$250=		0			
TOTAL ADD'L FEE						OR		TOTAL ADD'L FEE		0			
(Column 1)			(Column 2)			(Column 3)							
AMENDMENT	CLAIMS REMAINING AFTER AMENDMENT				HIGHEST NUMBER PREVIOUSLY PAID FOR		PRESENT EXTRA						
	Total (37 CFR 1.16(i))		*		Minus		**		=				
	Independent (37 CFR 1.16(h))		*		Minus		***		=				
	☐ Application Size Fee (37 CFR 1.16(s))												
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))												
RATE (\$)		ADDITIONAL FEE (\$)				RATE (\$)		ADDITIONAL FEE (\$)					
X \$ =						OR		X \$ =					
X \$ =						OR		X \$ =					
TOTAL ADD'L FEE						OR		TOTAL ADD'L FEE					

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.
 ** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".
 *** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".
 The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.

Legal Instrument Examiner:
/CAROL BARNES/

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